`kOmegaDynamic` RAS Turbulence Model

This document reports the implemented equations and the user-configurable parameters for the `kOmegaDynamic` turbulence model.

1 Mathematical Formulation

1.1 Transport Equations

The transport equations for the dynamic $k-\omega$ model with the zero-decay adjustment applied to the filtered variables (denoted by the subscript τ) are given by:

$$\frac{\partial k_\tau}{\partial t} + \frac{\partial k_\tau \langle u_j \rangle_\tau}{\partial x_j} = 2\nu_t \langle S_{ij} \rangle_\tau \langle S_{ij} \rangle_\tau - c_\mu^* \beta^* k_\tau \omega_\tau + \frac{\partial}{\partial x_j}\left[\left(\nu + \frac{\nu_t}{\sigma_k}\right)\frac{\partial k_\tau}{\partial x_j}\right] \tag{1}$$

$$\frac{\partial \omega_\tau}{\partial t} + \frac{\partial \omega_\tau \langle u_j \rangle_\tau}{\partial x_j} = 2\gamma \frac{\omega_\tau}{k_\tau} \nu_t \langle S_{ij} \rangle_\tau \langle S_{ij} \rangle_\tau - c_\mu^* \beta \omega_\tau^2 + \frac{\partial}{\partial x_j}\left[\left(\nu + \frac{\nu_t}{\sigma_\omega}\right)\frac{\partial \omega_\tau}{\partial x_j}\right] \tag{2}$$

Note on implementation: In the OpenFOAM source code, the turbulent Prandtl numbers σ_k and σ_ω are implemented using their inverses: $\alpha_k = 1/\sigma_k$ and $\alpha_\omega = 1/\sigma_\omega$.

1.2 Dynamic Procedure and Eddy Viscosity

The turbulent eddy viscosity is defined as:

$$\nu_t = c_\mu \frac{k_\tau}{\beta^* \omega_\tau} \tag{3}$$

The dynamic coefficient c_μ, which is bounded between $0 \leq c_\mu \leq 0.2$, is evaluated using the Germano-like dynamic procedure:

$$c_\mu = \frac{\mathcal{M}_{ij}\mathcal{L}_{ij}}{\mathcal{M}_{ij}\mathcal{M}_{ij}} \tag{4}$$

with the resolved stress tensor $\mathcal{L}_{ij}$ and model tensor $\mathcal{M}_{ij}$ defined over a time-averaging window $\langle \cdot \rangle_T$ as:

$$\mathcal{L}_{ij} = \langle \langle u_i \rangle_\tau \langle u_j \rangle_\tau \rangle_T - \langle \langle u_i \rangle_\tau \rangle_T \langle \langle u_j \rangle_\tau \rangle_T \tag{5}$$

$$\mathcal{M}_{ij}^{k-\omega} = 2\left\langle \frac{k_\tau}{\beta^* \omega_\tau} \langle S_{ij} \rangle_\tau \right\rangle_T - 2\frac{k_{T_\tau}}{\beta^* \omega_{T_\tau}} \langle \langle S_{ij} \rangle_\tau \rangle_T \tag{6}$$

1.3 Zero-Decay Modification

The normalized dynamic coefficient c_μ^* is introduced in the destruction terms of Eqs. 1 and 2. It is normalized with respect to its conventional value $c_{\mu,0} = 0.09$:

$$c_\mu^* = \frac{c_\mu}{c_{\mu,0}} \tag{7}$$

When the `Zero-Decay` switch is disabled in OpenFOAM, c_μ^* defaults to 1.0, reverting to the classical free-stream decay formulation.

2 OpenFOAM Configuration

To utilize the `kOmegaDynamic` model, the `momentumTransport` dictionary located in the `constant` directory must be configured as follows.

```
/*--------------------------------*- C++ -*----------------------------------*\
  =========                 |
  \\      /  F ield         | OpenFOAM: The Open Source CFD Toolbox
   \\    /   O peration     | Website:  https://openfoam.org
    \\  /    A nd           | Version:  10
     \\/     M anipulation  |
\*---------------------------------------------------------------------------*/
FoamFile
{
    format      ascii;
    class       dictionary;
    location    "constant";
    object      momentumTransport;
}
// * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * //
simulationType RAS;
RAS
{
    RASModel        kOmegaDynamic;
    turbulence      on;
    printCoeffs     on;
    kOmegaDynamicCoeffs
    {
        // Standard k-omega parameters
        betaStar        0.09;
        beta            0.075;
        gamma           0.55;
        alphaK          0.5;
        alphaOmega      0.5;

        // Dynamic procedure parameters
        Cmu_0           0.09;       // Base reference value
        dynamicCmuMin   0.0;        // Lower clipping limit
        dynamicCmuMax   0.2;        // Upper clipping limit

        // Averaging and stabilization
        nWindow         2;          // Buffer size for temporal averaging
        nStart          10;         // Time steps before dynamic procedure

        // Model variant switch
        Zero-Decay      yes;        // Activate Zero-Decay modification
    }
}
// ************************************************************************* //
```

constant/momentumTransport

2.1 Initial Conditions: `0/dynamicCmu`

Since c_μ is a calculated field, it must be initialized in the `0/` directory as a `volScalarField`. An example is reported in the following.

```
/*--------------------------------*- C++ -*----------------------------------*\
  =========                 |
  \\      /  F ield         | OpenFOAM: The Open Source CFD Toolbox
   \\    /   O peration     | Website:  https://openfoam.org
    \\  /    A nd           | Version:  10
     \\/     M anipulation  |
\*---------------------------------------------------------------------------*/
FoamFile
{
    format      ascii;
    class       volScalarField;
    location    "0";
    object      dynamicCmu;
}
// * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * //

dimensions      [0 0 0 0 0 0 0];
internalField   uniform 0.09;

boundaryField
{
    REAR
    {
        type            cyclicAMI;
        neighbourPatch  FRONT;
    }
    FRONT
    {
        type            cyclicAMI;
        neighbourPatch  REAR;
    }
    SIMETR
    {
        type            fixedValue;
        value           uniform 0;
    }
    WALL
    {
        type            fixedValue;
        value           uniform 0;
    }
    IN
    {
        type            fixedValue;
        value           uniform 0;
    }
    OUT
    {
        type            zeroGradient;
    }
    TOP
    {
        type            fixedValue;
        value           uniform 0;
    }
}
```

0/dynamicCmu

3 Parameter

The following table outlines the specific coefficients.

Parameter	Description	Default Value
β^* / `betaStar`	Constant for turbulence dissipation rate.	0.09
β / `beta`	Constant for specific dissipation destruction.	0.075
γ / `gamma`	Constant for specific dissipation production.	0.55
`alphaK`	Inverse turbulent Prandtl number for k ($1/\sigma_k$).	0.5
`alphaOmega`	Inverse turbulent Prandtl number for ω ($1/\sigma_\omega$).	0.5
`Cmu_0`	Base reference value for the C_μ coefficient.	0.09
`nWindow`	Number of previous time steps retained for temporal averaging.	2
`nStart`	Time steps before starting the dynamic procedure.	10
`Zero-Decay`	Boolean switch (`yes`/`no`) to activate the C_μ^*.	`yes`

Table 1: Specific parameters for the `kOmegaDynamic` model